Assignment 3 杜奕璇 122090095

1. (a) $\bar{x} = \frac{21+22+18+29+35+20+30+32+26+23+19+29+31+28+27}{15} = 26$

$s^2 = \frac{1}{14}\sum_{i=1}^{15}(x_i-\bar{x})^2 = 27.14$

(b) $H_0: \mu = \mu_0 = 26$

$H_1: \mu > \mu_0$

(c) $Z = \frac{\bar{x}-\mu_0}{\frac{6}{\sqrt{n}}} = 0$ $\quad Z_{0.05} \approx 1.645$

(d) Charles would not reject H_0

(e) $\frac{\bar{x}-26}{\frac{4}{\sqrt{15}}} > 1.645$ $\quad \bar{x} > 27.6989$

(f) when σ^2 is not known, $t = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}} = 0$

Charles would not reject H_0.

2. (a) $H_0: P_0 = \frac{1}{2}$, $H_1: P_0 \neq \frac{1}{2}$

$Z_{0.025} = 1.96$ $\quad \hat{P} = 0.6$ $\quad Z = \frac{\hat{P}-P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{0.6-0.5}{\sqrt{\frac{0.6\times0.4}{50}}} = 1.4434 < Z_{0.025}$

$\therefore$ Michael does not reject H_0

(b) $\hat{P} = \frac{h}{50}$, $Z = \frac{\frac{h}{50}-0.5}{\sqrt{\frac{\frac{h}{50}(1-\frac{h}{50})}{50}}} \geq 1.96$ $\quad h \geq 31.68$

or $Z = \frac{\frac{h}{50}-0.5}{\sqrt{\frac{\frac{h}{50}(1-\frac{h}{50})}{50}}} \leq -1.96$ $\quad h \leq 18.32$

$\therefore$ when $h \geq 32$ or $h \leq 18$, reject H_0.

3. (a) $\text{Power} = 1-\beta = P_r(\text{rejected } H_0 \mid H_0 \text{ false}) = P_r(\bar{x} > \mu_0 + Z_\alpha \frac{\sigma}{\sqrt{n}} \mid \mu = \mu')$

$= P_r(\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > Z_\alpha \mid \mu = \mu') = P_r(Z > \frac{\mu_0-\mu'}{\sigma/\sqrt{n}} + Z_\alpha \mid \mu = \mu')$

$= 1 - \Phi(Z_\alpha + \frac{\mu_0-\mu'}{\sigma/\sqrt{n}})$

$\therefore \beta(\mu') = 1 - \text{Power} = \Phi(Z_\alpha + \frac{\mu_0-\mu'}{\sigma/\sqrt{n}})$

(b) $\because 1-\beta = \Phi(Z_{1-\alpha} + (\mu_0-\mu')\sqrt{n}/\sigma)$ $\quad \therefore Z_{1-\alpha} + (\mu_0-\mu')\sqrt{n}/\sigma = Z_\beta$

$\therefore \sqrt{n} = \frac{(-Z_{1-\alpha}+Z_\beta)\sigma}{\mu_0-\mu'}$ $\quad \therefore n = \left[\frac{(Z_\alpha+Z_\beta)\sigma}{\mu_0-\mu'}\right]^2$

(c) $z_{0.025} = 1.96$ $z_{0.05} = 1.645$

$$n = \left(\frac{6(z_\alpha + z_\beta)}{\mu_0 - \mu'}\right)^2 = \left(\frac{1300(1.96+1.645)}{24000-25000}\right)^2 = 21.96$$

$\therefore n = 22$

4. (a) $H_0: P_A = P_B$, $H_1: P_A \neq P_B$

(b) $\hat{P_A} = 0.1205$ $\hat{P_B} = 0.1188$

(c) $Z = 1.3626$

P-value $= 0.1730$

(d) $\because$ p-value $= 0.1730 > 5\%$

$\therefore$ H_0 is not rejected at 5% significance level.

5. (a) $H_0: \mu_Y \geq \mu_X$

$H_1: \mu_Y < \mu_X$

(b) Assume $\sigma_X^2 = \sigma_Y^2$, $t = \dfrac{(\bar{X} - \bar{Y})}{\sqrt{\frac{S_p^2}{n} + \frac{S_p^2}{m}}}$ $S_p^2 = \dfrac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$

(c) Assume $\sigma_X^2 \neq \sigma_Y^2$, $t = \dfrac{(\bar{X} - \bar{Y})}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$

(d) $H_0: \sigma_X^2 = \sigma_Y^2$

$H_1: \sigma_X^2 \neq \sigma_Y^2$ $F = \dfrac{S_X^2}{S_Y^2}$

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(b) t_statistic is:-8.336873582326676  t_critical is:-1.644933572581743
 So reject H0: μX > μY
(c) t_statistic_welch is:-8.336873582326676  t_critical_welch is:-1.6449959837678225
So reject H0: μX > μY
(d) f_statistic is:1.0049224336665912 f_critical_low is:0.957221125506382 f_critical_high is:1.0451043846030437
Fail to reject H0
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